

Ultimate Nutrition

ARGININE PYROGLUTAMATE LYSINE

Growth hormone (GH) is central to growth and development. It is the hormone that allows growth through puberty into the adulthood. Its levels, however, begin to steadily decline after early adulthood such that by the sixth or seventh decade of life, GH must be supplemented to maintain good physical and mental health. It is not just the elderly who need GH replenishment. Athletes involved in intense exercise on a regular basis need boosting of GH to increase energy, enhance lean body mass and to maintain strong bones and keep their weight under control.

Ultimate Nutrition's Arginine Pyroglutamate Lysine is the blend of stacked amino acids that induces the secretion of GH. Both arginine and lysine have been taken in combination to boost the amounts of GH. Arginine pyroglutamate is the only form of arginine that, in combination with lysine, effectively increases GH levels orally. Availability of GH in systemic circulation also helps increase strength, burn fat, enhance libido, boost immunity and support tendons and ligaments. By inducing the secretion of GH from the pituitary, arginine and lysine stimulate muscle metabolism and, hence, provide nutritional support for the athlete.

Intense physical activity over longer periods of time can deprive the brain of energy necessary for its function. Taken orally, arginine pyroglutamate crosses into the brain, since it easily breaches the blood-brain barrier to stimulate energetic processes in the brain and improve cognitive functions. As such, Arginine Pyroglutamate Lysine is the perfect blend of amino acids that ensures proper nutrition for both the body and the mind.

Growth hormone is absolutely essential for optimal health at any stage, irrespective of chronological age or lifestyle choices, such as intense physical activity by and high-performance exercise. Maintaining ample supply of GH helps the body fend against the frailties that are a necessary result of the aging process. The beneficial effects of GH in the human body are global, and GH deficiency can cause an array of chronic conditions with advancing years. As such, Ultimate Nutrition's Arginine Pyroglutamate Lysine is both for active athletes as well as anyone motivated to maintain good health and enjoy the sense of general wellbeing well into the golden years of life.

SELECTED REFERENCES

1. Ahmed, A., Health Benefits of Human Growth Hormone, Keats Publishing, New Canaan, Connecticut, 1998
2. Ahmed, A., "Growth Hormone: Future Trends and Industry Perspective," In Anti-Aging Medical Therapeutics, (Klatz, R. and Goldman, R., Eds.), Health Quest Publication, Marina del Rey, California, 1997
3. Medina, J., The Clocks of Ages, Cambridge University Press, New York, New York, 1996
4. Toogood, A. O'Neill, P. and Shalet, S., "Beyond the Somatopause: Growth Hormone Deficiency in Adults Over the Age of 60 Years," J. Clin. Endocrinol. Metab.: 81, 460, 1996

5. Goth, M., Szabolcs, I. and Peter, F., "Growth Hormone Therapy in Adults," *Orev. Hetil.*: 136, 1243, 1995
6. Saini, S., Hindmarsh, P., Matthews, D., Pringle, P., Jones, J., Preece, M. and Brook, C., "Reproducibility of 24-Hour Serum Growth Hormone Profile in man," *Clin. Endocrinol.*: 34, 455, 1991
7. Isidori, A., Lo Monaco, A. and Cappa, M., "A Study of Growth Hormone Release in Man after Oral Administration of Amino Acids," *Curr. Med. Res. Opinion*: 7. 475, 1987
8. Kuhara, T., Ikeda, S., Ohneda, A. and sasaki, Y., "Effects of Intravenous Infusion of 17 Amino Acids on the Secretion of Growth Hormone, Glucagon and Insulin," *Am. J. Physiol.*: 260, E21, 1991

FREQUENTLY ASKED QUESTIONS

1) What Exactly is Arginine Pyroglutamate?

Arginine pyroglutamate is a substance in which the amino acid arginine is coupled with a pyroglutamate molecule. Arginine by itself has not been proven consistently in increasing the secretion of growth hormone or improvement of cognitive functions. Even though a few studies in the 1980s reported that arginine and lysine induced GH secretion, their independent verification by other investigators has not thus far been forthcoming. In contrast, arginine pyroglutamate and lysine have been shown to effectively increase GH levels in the body.

2) What Are the Dietary Sources of Arginine Pyroglutamate?

Pyroglutamate is a naturally occurring amino acid found in vegetables, fruit, dairy products and meat. The dietary intake may, however, not be sufficient for athletes or aging individuals whose needs for the secretion of GH is greater compared to those, respectively, with a more sedentary lifestyle or younger people.

3) Are There Any Side Effects to Arginine Pyroglutamate and Lysine?

No! These are amino acids that are normally present in the foodstuffs consumed by people on a daily basis. Of course, any adverse effects depend upon the amount of amino acids taken. If taken in the recommended dosage of one and one-half to two grams of arginine pyroglutamate and 750 to 1000 milligram of lysine, no side effects should result. It should be noted that lysine and arginine, as opposed to arginine pyroglutamate, have been reported to induce GH, albeit in amounts ranging from 15 to 30 grams. That is too large an amount, and in such quantities amino acids may well cause dizziness, nausea and frothing at the mouth. Ultimate Nutrition's Arginine Pyroglutamate/Lysine combines these amino acids in the right amounts ingested over a 24-hour period to synchronize GH secretion with the body's diurnal rhythms.

4) What is the Best Time to Take Arginine Pyroglutamate Lysine?

It is best taken before bedtime to maximize its beneficial effects and to allow GH secretion to synchronize with the body's own rhythm of GH production.